

Direct Preference Optimization

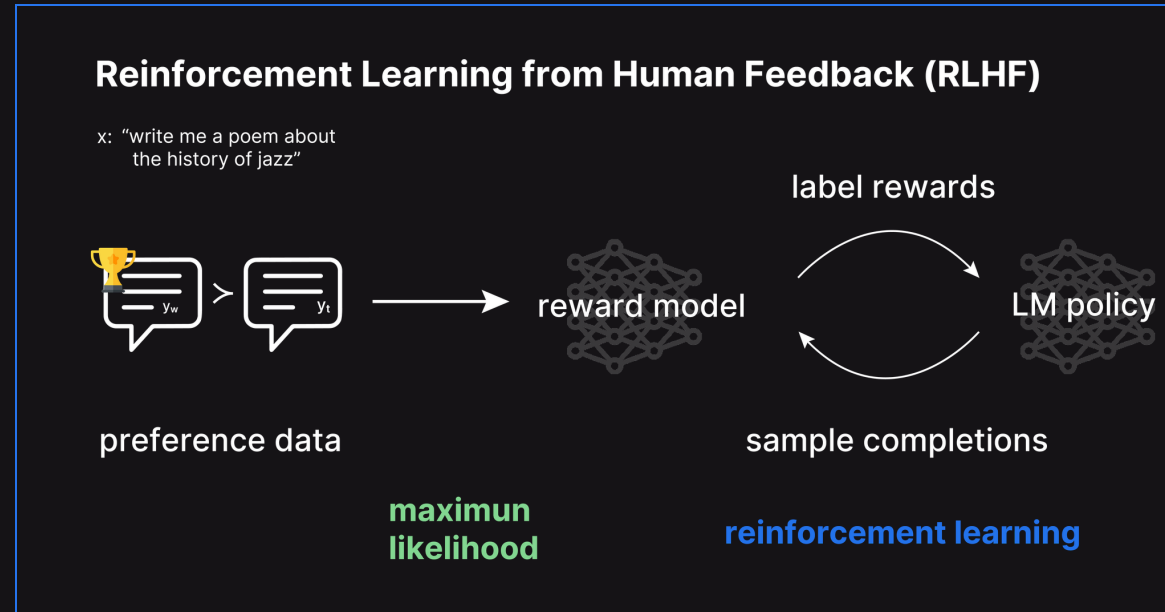
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Drawbacks of RLHF

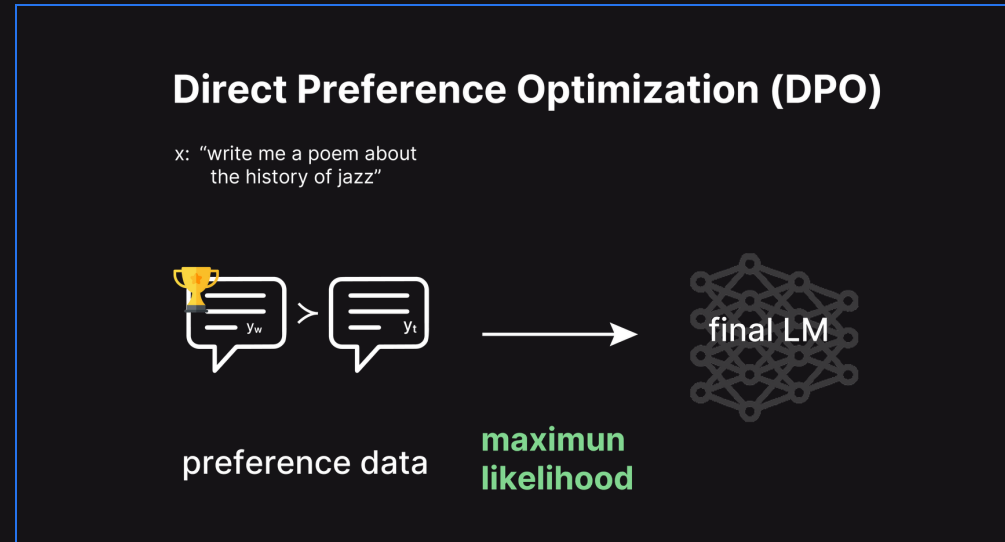


Complexity: Designing and training a reward model is challenging and demands expertise in RL

Computational Overhead: The process of training both the LLM and the reward model requires substantial computational resources

Reduce Control: Users have less direct influence over the LLM's behaviour since the reward model intermediates the feedback process.

Direct Preference Optimization



Optimization Process : Instead of using a reward model, DPO directly integrates these preferences into the model's loss function, optimizing the model to produce outputs that aligns with user preferences

Gradient Descent: The model parameters are updated using gradient descent based on the user preferences data, guiding the model towards behaviours that reflect the desired outcomes

No Reward Model: Unlike RLHF, DPO skips the creation and training of a reward model, streamlining the process and focusing purely on aligning the model with

Direct Feedback Incorporation: User feedback is directly applied during training, making the optimization process more intuitive and easier to interpret

Direct Preference Optimization

Simplicity : DPO is easy to implement, with no need for complex reward models, making it accessible even to those without deep RL expertise

Efficiency: By removing the reward model, DPO significantly cuts computational costs, ideal for large-scale deployments

Control: Users have more direct influence over LLM behaviour, allowing precise alignment with their goals

Faster Convergence: DPO's straightforward approach often leads to quicker results, beneficial for tasks needing rapid iteration

Improved Performance: Research shows DPO can outperform RLHF in tasks like sentiment control and dialogue, making it a strong fine-tuning option

Thank You
